AudioConditioner

Models & Training

Keyvan Attarian & Baptiste Arnaudo

CSC_52002_EP - Multimodal generative AI

Introduction & The Descriptor Model

AudioConditioner is a high-level multimodal model that we developed to bridge the gap between visual/textual semantics and music generation. We utilize BLIP for image captioning and the CLAP text encoder for semantic embeddings.

The central component is our **Descriptor Model**, which is the only trainable component in the entire architecture. We keep all other models (BLIP, CLAP, Stable Audio) completely frozen. The Descriptor acts as a multi-task learning bridge; we map the CLAP text embedding through a shared neural backbone into 12 discrete and continuous musical attributes.

We structurally combine the generated attributes within a **MusicDescriptor** object. We then use this object to programmatically generate the descriptive text prompt fed into Stable Audio (which has a 47-second generation limit). Finally, to ensure semantic alignment, we generate N candidates and re-rank them using cosine similarity against the CLAP text embedding.

The MusicDescriptor Schema

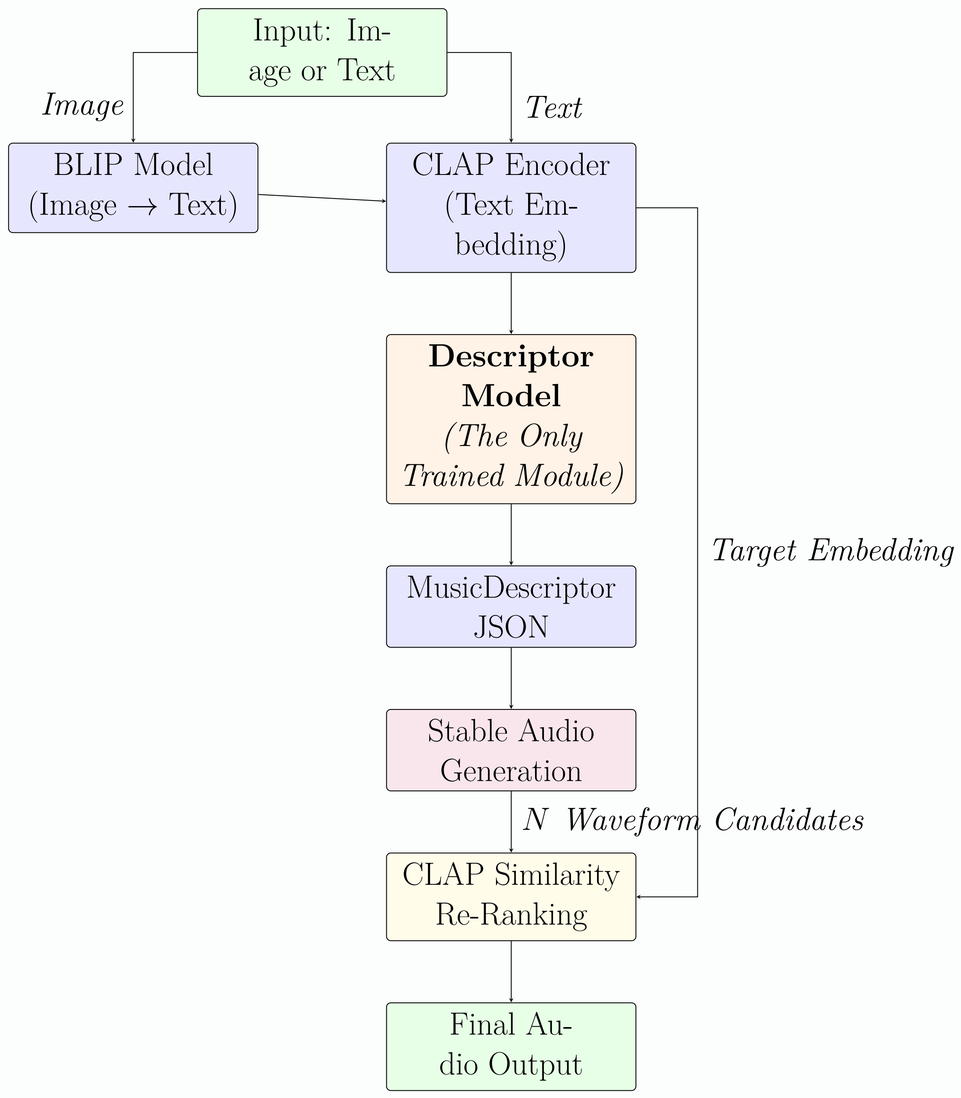
Classification Attributes

- **Mood** ($\rightarrow$ Multi-Label List): Emotional states (e.g., dark, epic, melancholic).
- **Key/Mode** ($\rightarrow$ Single-Label): The musical scale used (e.g., major, minor).
- **Instrumentation** ($\rightarrow$ Multi-Label List): Active instruments (e.g., strings, synth).
- **Rhythm Style** ($\rightarrow$ Single-Label): Overall rhythmic feel (e.g., sparse, pulsing).
- **Structure** ($\rightarrow$ Single-Label): Internal song progression (e.g., slow-build).
- **Production Style** ($\rightarrow$ Multi-Label List): Mixing/Genre aesthetic (e.g., cinematic).
- **Dynamics Profile** ($\rightarrow$ Single-Label): Volume progression (e.g., gradual build).

Regression Attributes

- **Energy** ($[0.0, 1.0]$): Intensity level.
- **Valence** ($[0.0, 1.0]$): Emotional polarity (0.0 = sad, 1.0 = happy).
- **Tempo** ($[50, 180]$ BPM): Beats per minute, normalized to $[0.0, 1.0]$.
- **Harmonic Tension** ($[0.0, 1.0]$): Chord dissonance.
- **Texture Density** ($[0.0, 1.0]$): Number of simultaneous elements.

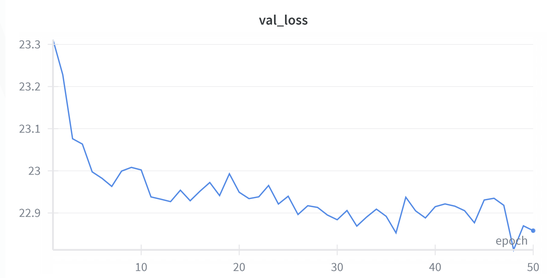
Framework Architecture

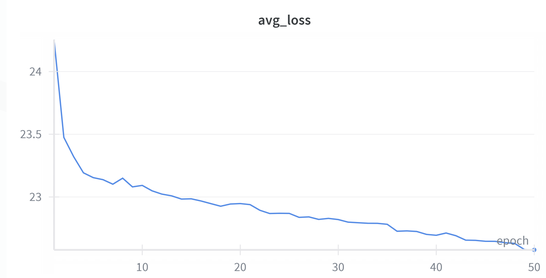


Phase 1: Generative LLM Distillation

We trained the **Descriptor Model** using a parallel corpus mapping natural language to the strict **MusicDescriptor** format. We built this dataset through prompt mixing and teacher LLM distillation:

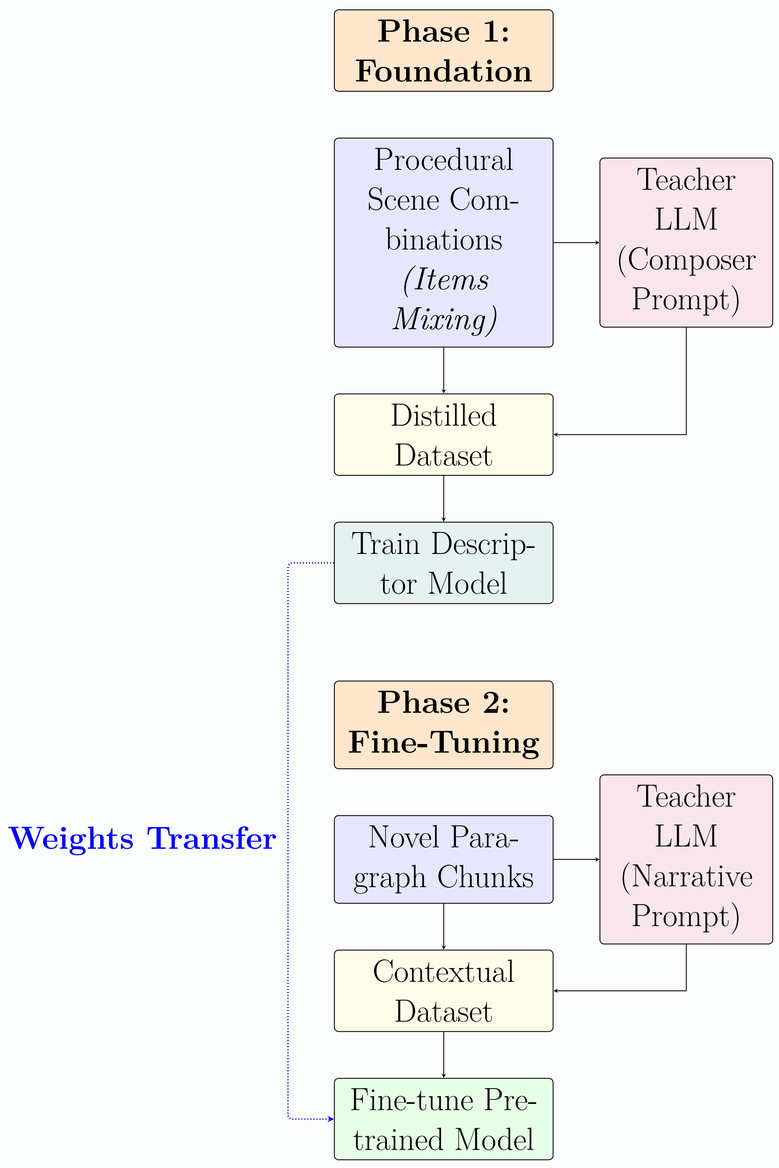
- **Data Generation:** We procedurally generated short scene combinations by randomly mixing predefined characters, settings, and events (e.g., *"A lonely astronaut under a sunset discovers an opportunity."*).
- **Distillation:** We prompted an LLM to act as a composer, taking these scenes and outputting the deterministic JSON MusicDescriptor fitting each scene.
- **Optimization:** We trained the model using an AdaptedMusicDescriptorLoss function (Cross-Entropy for categorical attributes and MSE for continuous target regressions).





Phase 2: Contextual Novel Fine-Tuning

To support narrated audiobooks with adaptive background music (ReadMusic), we fine-tuned the model to handle longer contextual evolution:

1. **Novel Chunking:** We split public domain novels into consecutive paragraphs (up to 80 words).
2. **Target Generation:** We used the teacher LLM to assign progressive MusicDescriptors respecting the storyline's emotional flow.
3. **Fine-Tuning Outcomes:** The descriptor internalizes literary context; we aim to prevent abrupt changes and enable continuous emotional transitions for long-form reading.

Training Pipeline Overview